benchmarking tools ENCODE data correlation for checking sample mate

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Contents

1 How to generate correlation input data

```
# option 1
#Step1: for each sample
#samtools mpileup -f /cfs/klemming/home/q/qunli/qunl/Qun/publicData/reference/hg38/addChr/bwaIndex/hg38
#java -jar /cfs/klemming/home/q/qunli/Software/benchmarkingTools/VarScan.v2.4.6.jar mpileup2snp check/A
#bgzip ATAC.vcf
#bcftools index ATAC.vcf.gz
#bcftools reheader -s <(echo "Sample1_ATAC") ATAC.vcf.gz -o ATAC_renamed.vcf.gz
#bcftools index ATAC_renamed.vcf.gz

#Step2: merge all samples and extract freq
#bcftools merge -Oz --force-samples -o merged_samples.vcf.gz ATAC_renamed.vcf.gz RNA_renamed.vcf.gz snA
#bcftools query -f '%CHROM\t%POS[\t%FREQ]\n' merged_samples.vcf.gz > merged_samples_freq.vcf

# option 2
#for sample in ENCFF260XKB ENCFF413SRZ ENCFF422XWI ENCFF455YNB ENCFF491HQL ENCFF692LTO ENCFF354SCV ENCF
#do
#  java -jar /cfs/klemming/home/q/qunli/Private/scmutrace/software/picard.jar AddOrReplaceReadGroups \
#    I=${sample}.bam \
#    O=${sample}.RG.bam \
#    RGID=${sample} \
#    RGLB=lib1 \
```

```
#    RGPL=ILLUMINA \
#    RGPU=unit1 \
#    RGSM=${sample}

#  samtools index ${sample}.RG.bam
#done


#java -jar /cfs/klemming/home/q/qunli/Private/scmutrace/software/picard.jar CrosscheckFingerprints \
#  INPUT=ENCFF260XKB.RG.bam \
#  INPUT=ENCFF422XWI.RG.bam \
#  INPUT=ENCFF413SRZ.RG.bam \
#  INPUT=ENCFF692LTO.RG.bam \
#  INPUT=ENCFF491HQL.RG.bam \
#  INPUT=ENCFF455YNB.RG.bam \
#  INPUT=ENCFF354SCV.RG.bam \
#  INPUT=ENCFF593YGT.RG.bam \
#  HAPLOTYPE_MAP=Homo_sapiens_assembly38.haplotype_database.txt \
#  CROSSCHECK_BY=SAMPLE \
#  LOD_THRESHOLD=-5 \
#  EXPECT_ALL_GROUPS_TO_MATCH=true \
#  OUTPUT=sample.crosscheck_metrics
```

2 ENV

```
rm(list = ls())
setwd("/Users/liqun/Desktop/scMutrace-tutorial/ReproducibilityAnalysis/Figure2/")
```

3 libraries

```
library(pheatmap)
```

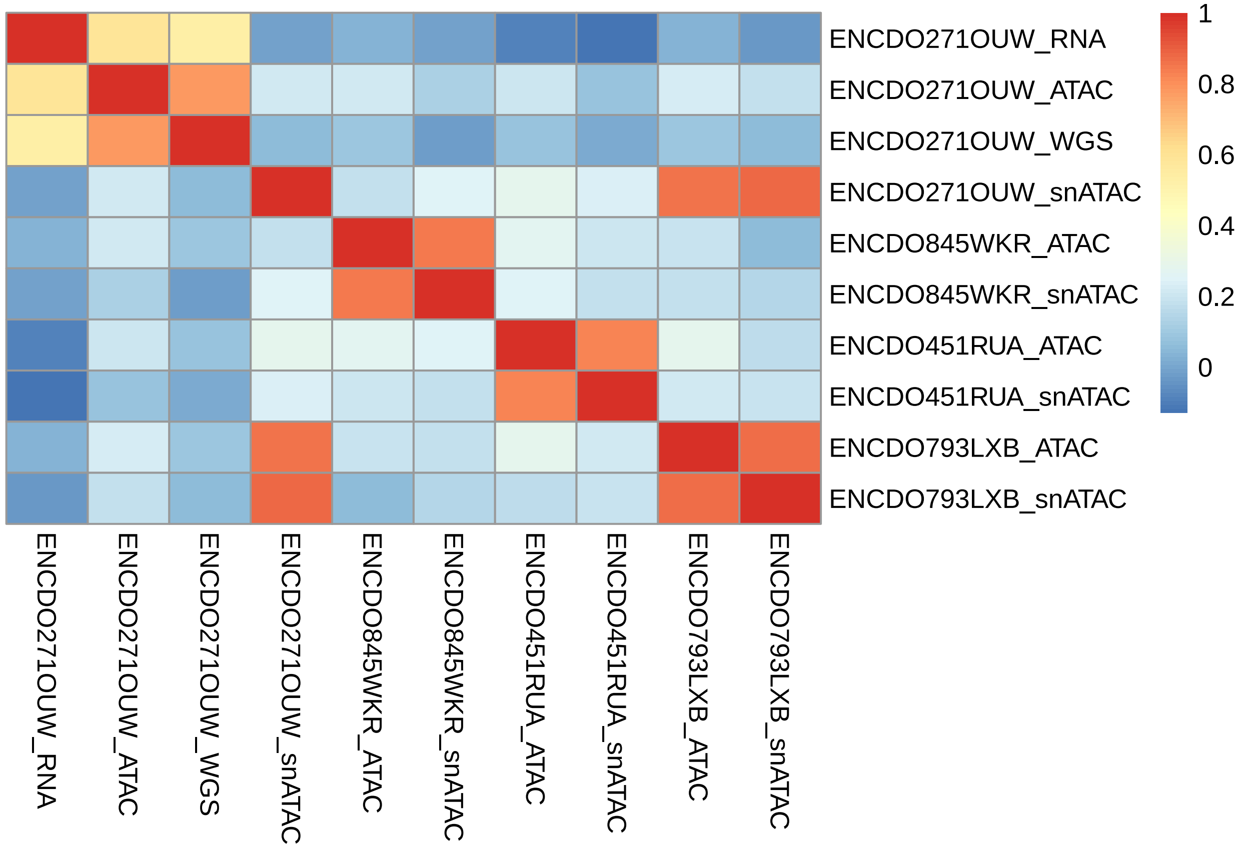
4 Load data

```
VAF <- read.table("../Data/HumanColon/ENCODE_merged_samples_freq.vcf", sep = "\t")
colnames(VAF) <- c("chr", "pos", "ENCD0271OUW_ATAC", "ENCD0271OUW_RNA", "ENCD0271OUW_snATAC", "ENCD0271
                    "ENCD0845WKR_ATAC", "ENCD0845WKR_snATAC",
                    "ENCD0451RUA_ATAC", "ENCD0451RUA_snATAC",
                    "ENCD0793LXB_ATAC", "ENCD0793LXB_snATAC")
VAF_frq <- VAF[,c("ENCD0271OUW_RNA", "ENCD0271OUW_ATAC", "ENCD0271OUW_WGS", "ENCD0271OUW_snATAC",
                   "ENCD0845WKR_ATAC", "ENCD0845WKR_snATAC",
                   "ENCD0451RUA_ATAC", "ENCD0451RUA_snATAC",
                   "ENCD0793LXB_ATAC", "ENCD0793LXB_snATAC")]

vaf_num <- VAF_frq
```

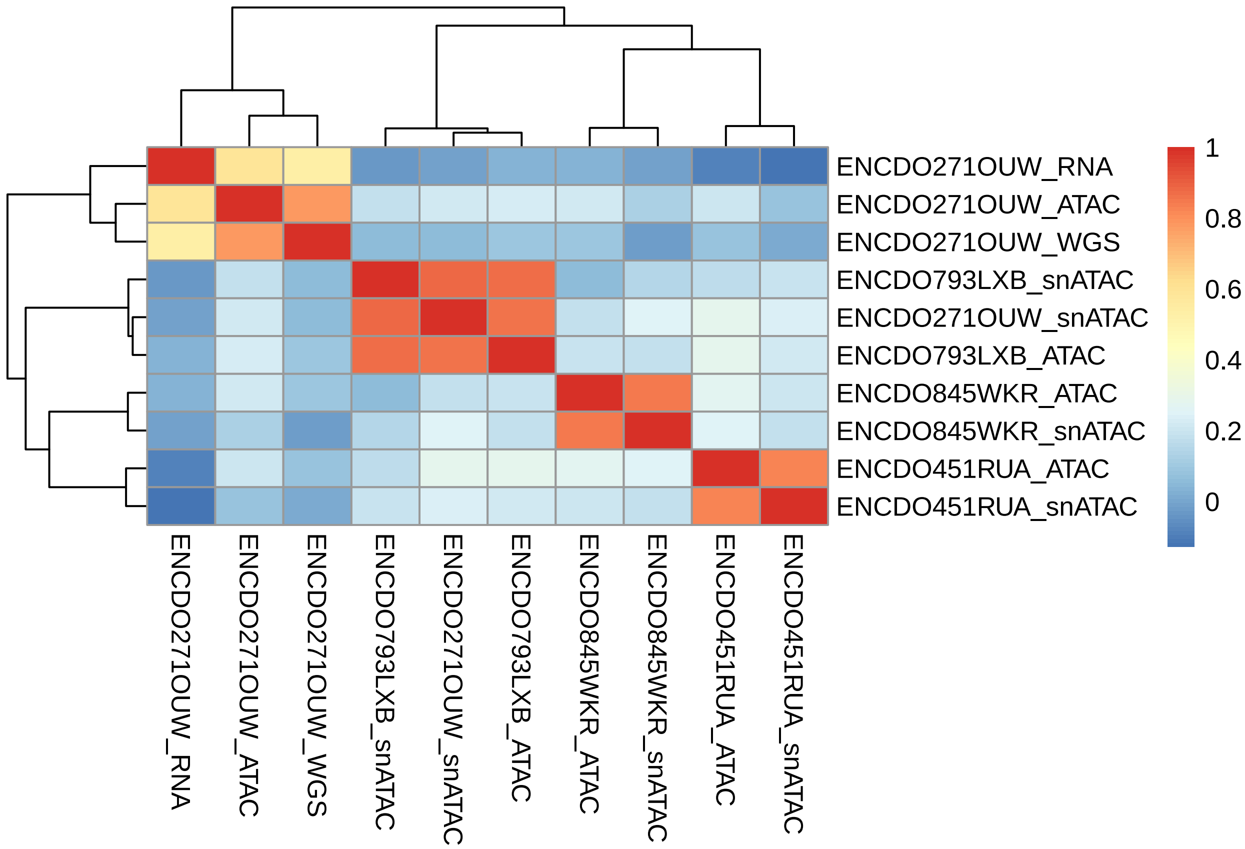
```r
vaf_num <- apply(vaf_num, 2, function(x) {
  x <- gsub("%","", x)
  x[x=="."] <- NA
  as.numeric(x)
})

cor_matrix <- cor(vaf_num, use="pairwise.complete.obs", method="pearson")
```

5 plot correlation

```r
pheatmap(cor_matrix, cluster_rows = F, cluster_cols = F)
```




```r
pheatmap(cor_matrix)
```



SNP-based allele fraction correlation analysis revealed that sample ENCDO271OUW_snATAC showed high genetic concordance with ENCDO793LXB, suggesting a potential sample mislabeling event.